11. Using interpolation estimate the business done in 1985 from the following data

Year	1982	1983	1984	1986
Business done (in lakhs)	150	235	365	525

12. Using interpolation, find the value of $f(x)$ when $x = 15$

x	3	7	11	19
$f(x)$	42	43	47	60

Exercise 5.3

Choose the correct Answer

1. $\Delta^2 y_0 =$

   (a) $y_2 - 2y_1 + y_0$ (b) $y_2 + 2y_1 - y_0$

   (c) $y_2 + 2y_1 + y_0$ (d) $y_2 + y_1 + 2y_0$

2. $\Delta f(x) =$

   (a) $f(x+h)$ (b) $f(x) - f(x+h)$

   (c) $f(x+h) - f(x)$ (d) $f(x) - f(x-h)$

3. $E \equiv$

   (a) $1+\Delta$

   (b) $1-\Delta$

   (c) $1+\nabla$

   (d) $1-\nabla$

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4. If h=1, then $\Delta(x^2) =$

   (a) $2x$ (b) $2x-1$

   (c) $2x+1$ (d) 1

5. If c is a constant then $\Delta c =$

   (a) c (b) Δ

   (c) Δ^2 (d) 0

6. If m and n are positive integers then $\Delta^m \Delta^n f(x) =$

   (a) $\Delta^{m+n} f(x)$ (b) $\Delta^m f(x)$

   (c) $\Delta^n f(x)$ (d) $\Delta^{m-n} f(x)$

7. If 'n' is a positive integer $\Delta^n \left[\Delta^{-n} f(x)\right]$

   (a) $f(2x)$ (b) $f(x+h)$

   (c) $f(x)$ (d) $\Delta f(x)$

8. $E f(x) =$

   (a) $f(x-h)$ (b) $f(x)$

   (c) $f(x+h)$ (d) $f(x+2h)$

9. $\nabla \equiv$

   (a) 1+E (b) 1-E

   (c) $1 - E^{-1}$ (d) $1 + E^{-1}$

10. $\nabla f(a) =$

    (a) $f(a) + f(a-h)$

    (b) $f(a) - f(a+h)$

    (c) $f(a) - f(a-h)$

    (d) $f(a)$

11. For the given points (x_0, y_0) and (x_1, y_1) the Lagrange's formula is

    (a) $y(x) = \frac{x-x_1}{x_0-x_1} y_0 + \frac{x-x_0}{x_1-x_0} y_1$

    (b) $y(x) = \frac{x_1-x}{x_0-x_1} y_0 + \frac{x-x_0}{x_1-x_0} y_1$

    (c) $y(x) = \frac{x-x_1}{x_0-x_1} y_1 + \frac{x-x_0}{x_1-x_0} y_0$

    (d) $y(x) = \frac{x_1-x}{x_0-x_1} y_1 + \frac{x-x_0}{x_1-x_0} y_0$

12. Lagrange's interpolation formula can be used for

    (a) equal intervals only

    (b) unequal intervals only

    (c) both equal and unequal intervals

    (d) none of these.

13. If $f(x) = x^2 + 2x + 2$ and the interval of differencing is unity then $\Delta f(x)$

    (a) $2x-3$ (b) $2x+3$

    (c) $x+3$ (d) $x-3$

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